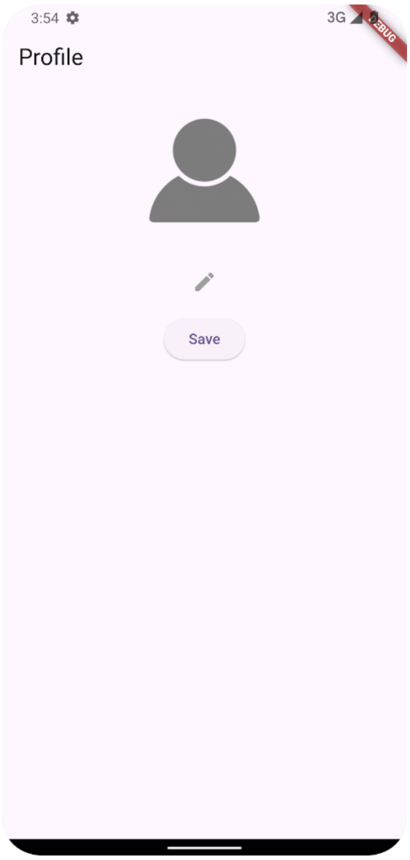
Practical 8: Data File

This app allows users to pick an image from the Gallery and save it as a profile picture in a folder dedicated to the app. This app utilizes the image picker and file Input/Output operations.

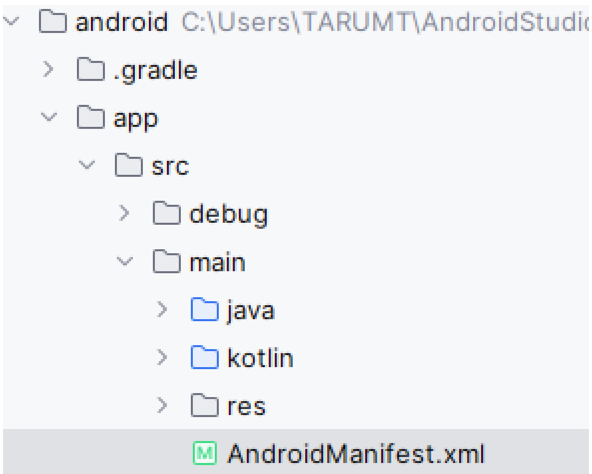
1. Create a new Flutter app using the Application template.



Default profile picture

Note: Download a profile image and insert it to the assets folder of your app.

2. In the project File Explorer, open the Insert the following permission into the Android'smanifest file.



```
<manifest xmlns:android="http://schemas.android.com/apk/res/android">
    <uses-permission android:name="android.permission.WRITE_EXTERNAL_STORAGE"/>
    <application
        android:label="helloflutter"
        android:name="${applicationName}"
        android:icon="@mipmap/ic_launcher">
```

```
<uses-permission android:name="android.permission.WRITE_EXTERNAL_STORAGE"/>
```

3. In the terminal, run the following command

```
flutter pub add image_picker path_provider
```

4. Open the **main.dart** file. Import the following packages:

```
import 'package:image_picker/image_picker.dart';
import 'package:path_provider/path_provider.dart';
```

5. Within the stateful widget, declare a file and an image picker instance:

```
//TODO - Create a file and an image picker instances
File? _image;
final picker = ImagePicker();
```

6. Create a method to obtain an image file from the Gallery:

```
//TODO - Obtain profile image from the Gallery
Future getImageFromGallery() async {
  final pickedFile = await picker.pickImage(source: ImageSource.gallery);

  if (pickedFile != null) {
    setState(() {
      _image = File(pickedFile.path);
    });
  }
}
```

7. Create a method to save the profile image into the app folder:

```
//TODO - Save the profile image into the app folder
Future<void> savePicture() async {
  if (_image != null) {
    try {
      // Save the image to a specific location
      final appDocDir = await getApplicationDocumentsDirectory();
      final newImagePath = '${appDocDir.path}/profile.png';
      await _image!.copy(newImagePath);
      print('File image copied successfully to $newImagePath');
    } catch (e) {
      print('File error copying image: $e');
    }
  } else {
    AlertDialog(
      title: const Text('Profile Image'),
      content: const SingleChildScrollView(
        child: ListBody(
          children: <Widget>[
            Text('Profile Image'),
            Text('Profile Image file is missing.'),
```

```
          ],
        ),
      ),
      actions: <Widget>[
        TextButton(
          child: const Text('Close'),
          onPressed: () {
            Navigator.of(context).pop();
          },
        ),
      ],
    );
  }
}
```

8. Create a method to load the profile image:

```
//TODO - Load profile image from the app folder
Future<void> loadProfileImage() async {
  // Get the application documents directory
  final appDocDir = await getApplicationDocumentsDirectory();
  final imagePath = '${appDocDir.path}/profile.png';

  // Create the destination file path
  final file = File(imagePath);

  if (await file.exists()) {
    setState(() {
      _image = file;
      print('File path: $imagePath');
    });
  } else {
    print('File not found in $imagePath');
  }
}
```

9. Call the loadProfileImage method in the initState method:

```
@override
void initState() {
  super.initState();
  //TODO - Call the LoadProfileImage method during app initialization stage
  loadProfileImage();
}
```

10. Construct the main UI:

```
//TODO - Construct the main UI
_image == null
    ? Image.asset(
        'assets/profile.png',
        width: 150,
        height: 150,
      )
    : Image.file(
        _image!,
        width: 150,
        height: 150,
      ),
const SizedBox(height: 8.0,),
IconButton(
```

```
      icon: const Icon(Icons.edit),
      color: Colors.grey,
      onPressed: () {
        getImageFromGallery();
      },
    ),
    const SizedBox(height: 8.0,),
    ElevatedButton(
        onPressed: () {
          savePicture();
          ScaffoldMessenger.of(context).showSnackBar(
              const SnackBar(content: Text('Profile image saved.')));
        },
        child: const Text('Save'),
    ),
```

11. Run the app. Pick an image from the Gallery and hit the Save button.

12. Click the menu **View** -> **Tool Windows** -> **Device Explorer**. Expand the folder data -> data >- <your_app_package_name> -> app_flutter, you should be able to find the image file named profile.png. Congratulations! You have completed this practical.

TODO: Enhance the app by reducing the image size – apply an image compression library.